

Betsy: Autonomous Procurement Agent

Before & After Business Process Model showing the operational impact of an Agentic AI tool on a small manufacturing procurement workflow. Grounded in the Betsy mock environment (12 SKUs, 6 suppliers, 4 live scenarios).

Project: Betsy — Autonomous Procurement POC**Primary Stakeholder:** Jenny (Operations Manager)**Date:** 28 May 2026**Phase:** Phase 2 — Core Autonomous Loop

Executive Summary

Operations Manager Jenny currently spends **30-35 hours per week** on procurement firefighting — manually checking inventory, chasing supplier quotes by email, drafting purchase orders, following up on deliveries, and reconciling invoices by hand. This reactive, sequential workflow leaves no time for strategic work and creates persistent risk of production-line shutdowns and duplicate payments.

- The Betsy agent **automates ~80% of procurement touchpoints**, running the detect → source → order → reconcile cycle continuously and without human initiation.
- Human review is preserved and meaningful — scoped to high-value decisions, price anomalies, and flagged exceptions — reducing Jenny's procurement load to **2-4 hours per week**.
- Three live mock scenarios demonstrate this concretely: Betsy detected a **1.5-day stockout window** for SKU-003 Steel Rods and dispatched a PO to QuickShip Express before any human noticed; identified a **\$1,470 duplicate invoice** (INV-2024-099 matching INV-2024-089) in under 100ms; and correctly escalated a **50% price spike** on Steel Rods rather than auto-purchasing.

Key Impact Metrics

WEEKLY PROCUREMENT TIME (JENNY)

Before: 30-35 hrs / week

2-4 hrs

After: Exception oversight only

~87% reduction

TASK AUTOMATION RATE

Before: 0% automated

~80%

After: Handled autonomously

Project plan target

SUPPLIER SOURCING LATENCY

Before: 24-48 hrs (email)

<5 sec

After: All 6 suppliers, instant

>17,000x faster

INVOICE DUPLICATE DETECTION

Before: Manual, unreliable

100%

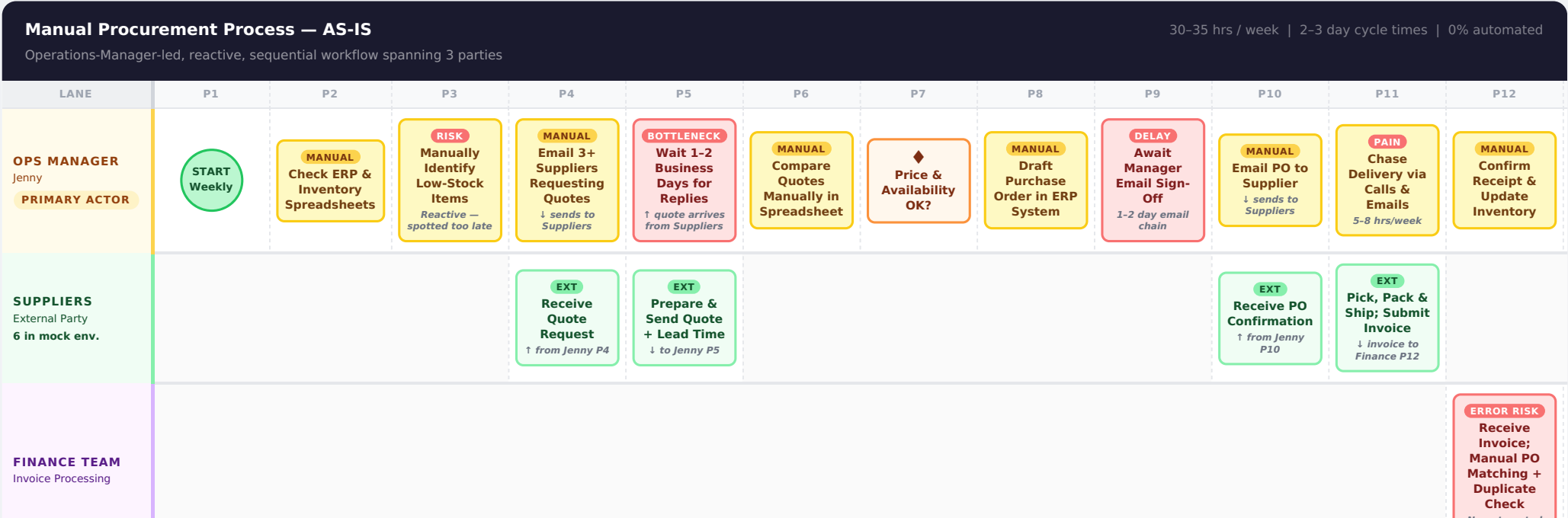
After: Every invoice scanned

Caught: \$1,470 dup. invoice

Manual Task (human effort) Automated Task (Betsy) Key Automated Step (high impact) Delay / Bottleneck / Risk External Party (Supplier) Decision Gateway Finance Activity Start Event End Event

AS-IS Process Model BEFORE BETSY

Current-state manual procurement workflow. Jenny is the primary actor across all phases — from inventory monitoring through invoice reconciliation. Key pain points are highlighted. Read left-to-right; vertical cross-lane references (↓ / ↑) indicate handoffs between parties.



No automated detection

Key AS-IS Pain Points

P3 — Reactive Detection
Stockouts identified only during Jenny's periodic manual checks. SKU-003 Steel Rods reached 45 units (1.5 days of production cover) before being noticed. Reorder point: 200 units.

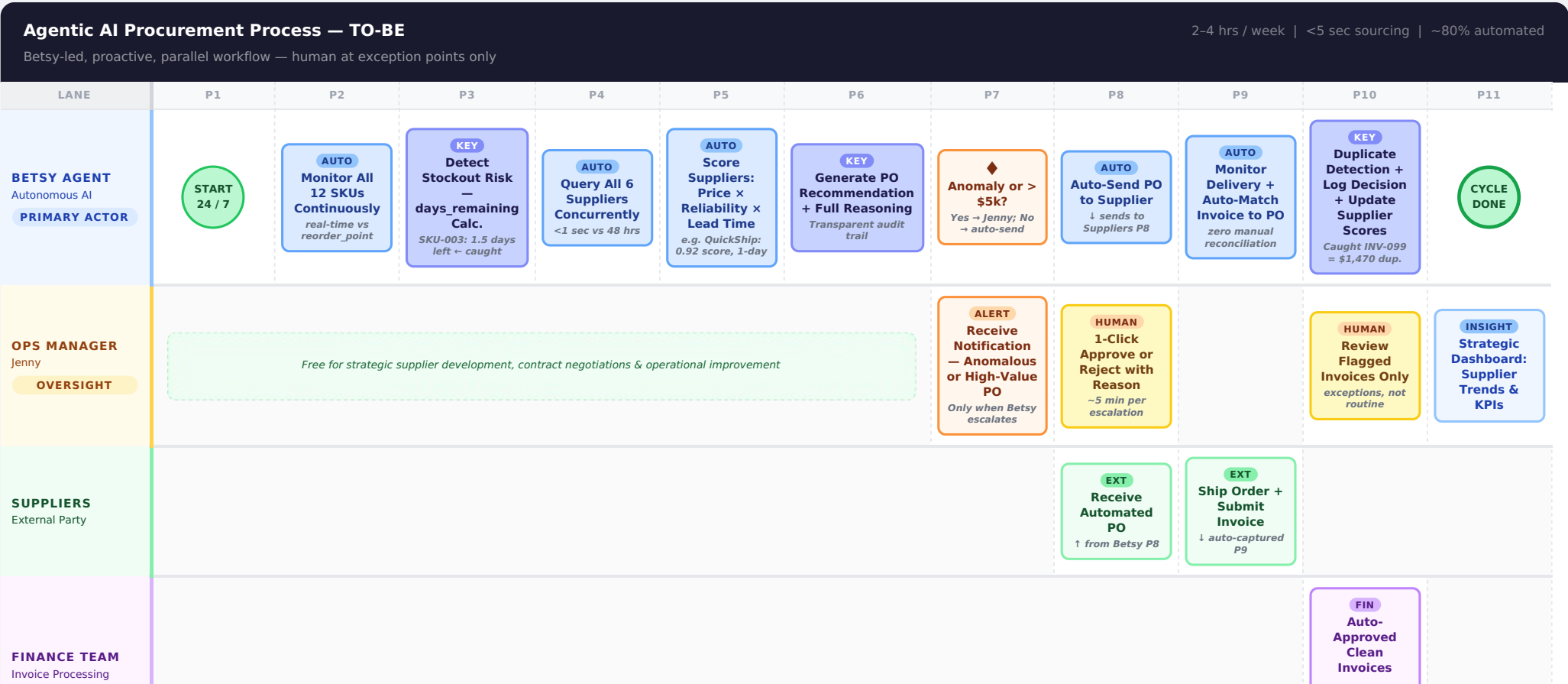
P5 — 48-Hour Sourcing Bottleneck
Email-based RFQ with 3–4 suppliers takes 1–2 business days. Not all 6 suppliers are contacted every cycle. QuickShip Express (1-day lead — best for urgent orders) often missed.

P9 — Approval Chain Latency
Manager sign-off via email adds 1–2 days. Weekends, holidays, and meetings extend this further — compounding the stockout risk window for critical SKUs.

P12 — Invoice Errors Undetected
Manual duplicate detection fails at scale. INV-2024-089 and INV-2024-091 are identical \$1,470 invoices from MidRange Supply (SUP-006) for 10× Circuit Board Type A — both in the system with no flag.

TO-BE Process Model AFTER BETSY

Future-state agentic AI-augmented procurement workflow. Betsy autonomously handles the full detect-source-order-reconcile loop. Jenny is reserved for high-value decisions, anomalies, and strategic oversight. The shorter diagram reflects fewer steps — a visual representation of the efficiency gain.



Key TO-BE Process Gains

- P3 — Proactive Detection**
Betsy calculates `days_remaining = current_stock / daily_usage_avg` continuously. SKU-003: $45 \div 30 = 1.5$ days. PO triggered immediately — no waiting until Monday's manual check.
- P4-P5 — Instant Parallel Sourcing**
All 6 suppliers queried simultaneously from the catalog in under 1 second. QuickShip Express (SUP-003, 1-day lead, 0.92 reliability) automatically selected for the urgent SKU-003 stockout.
- P7 — Smart Escalation Only**
Price spike scenario: SKU-003 best quote (\$13.60) is 19.3% above historical avg (\$11.40). At the 30% threshold, normal orders auto-approve; anomalies route to Jenny with full context and reasoning.
- P10 — Zero Duplicate Payments**
`_find_duplicates()` scans on every invoice submission. INV-2024-099 matched INV-2024-089: same SUP-006, same \$1,470, 29 days apart → HIGH RISK flag raised before any payment is processed.

Step-by-Step Process Comparison

Process Activity	AS-IS — Manual (Before)	TO-BE — Agentic AI (After)	Impact
Inventory Monitoring	Jenny manually checks ERP spreadsheet once or twice a week. Stockouts often identified with fewer than 2 days of cover remaining.	Betsy polls <code>/api/inventory</code> continuously. Calculates <code>days_remaining = current_stock ÷ daily_usage_avg</code> for all 12 SKUs. Threshold alert fires instantly.	Reactive weekly → proactive real-time. SKU-003 example: 1.5 days remaining detected immediately vs. noticed Monday morning after the fact.
Supplier Sourcing	Email 3-4 known suppliers. Wait 1-2 business days for replies. Not all 6 suppliers are contacted every cycle. Quotes compared manually in spreadsheet.	Betsy queries all 6 suppliers via <code>/api/suppliers</code> simultaneously. Filters by availability: true and catalog SKU match. Total time: under 1 second.	48-hour sourcing cycle → under 1 second. All 6 suppliers evaluated every time. Best-scoring option selected, not just whoever replies first.
Supplier Selection	Manual comparison based on experience and last known price. Reliability history not systematically tracked or weighted.	Multi-factor scoring across <code>unit_price</code> , <code>lead_days</code> , and <code>reliability_score</code> . For urgency: QuickShip (0.92 score, 1-day lead) beats BulkSupply (0.88, 7-day) despite higher price.	Gut-based → evidence-based. Supplier <code>reliability_score</code> updates after each transaction outcome (learning loop). Decision rationale logged for audit.
Purchase Order Creation	Jenny drafts PO manually in ERP. Sends manager an email for sign-off. Waits 1-2 days. Then manually emails PO to supplier. Entire cycle: 2-3 days.	Betsy POSTs to <code>/api/purchase-orders</code> with full reasoning. Status set to "pending_approval" for anomalies; auto-transitions to "approved" for routine orders. PO dispatched immediately.	2-3 day cycle → minutes for routine orders. Manager approval preserved for high-value/anomaly via push notification. Zero lag on standard reorders.
Delivery Tracking	Jenny calls or emails supplier to check status. Estimated 5-8 hours per week of follow-up. No systematic monitoring; surprises on Monday mornings.	Betsy monitors <code>expected_delivery</code> from PO record. PATCHes <code>/api/purchase-orders/{po_id}/status</code> on confirmation. Proactive late-delivery alert sent to Jenny if window is missed.	5-8 hrs/week chasing → zero manual effort. Proactive alerts replace reactive Monday morning discoveries.
Invoice Reconciliation	Finance manually matches invoice to PO in spreadsheet. Ad hoc duplicate check. INV-2024-089 and INV-2024-091 — both \$1,470 from SUP-006 — paid without a flag being raised.	Betsy runs <code>_find_duplicates()</code> on every invoice submission: matches <code>supplier_id</code> , <code>total_amount</code> , and date proximity (<60 days). HIGH RISK if under 30 days apart.	Manual / error-prone → automatic, 100% coverage. INV-2024-099 flagged immediately as duplicate of INV-2024-089 (same \$1,470, SUP-006, 29 days — HIGH RISK).
Price Anomaly Detection	No systematic baseline comparison. Jenny may sense	Betsy compares current quote against stored	No guardrail → enforced 30% threshold. Price

	"this seems high" but has no formal threshold. Risk of overpaying during supply disruptions under production pressure.	historical_avg_price. If deviation exceeds the 30% spike threshold, escalates to Jenny with all quotes, history, and recommendation rather than auto-ordering.	spike scenario (\$13.60 vs \$11.40 avg) correctly triggers flag_for_approval action before any PO is issued.
Decision Audit Trail	Ad hoc — email chains and some ERP entries. No structured log of why supplier X was chosen over Y. Finance cannot trace decision rationale for any given PO.	Every decision logged to /api/agent-log: timestamp, SKU, candidate suppliers with scores, chosen supplier, reasoning, and eventual outcome. Human-readable and queryable.	No systematic record → full transparency. Satisfies finance audit requirements. Enables bias review. Every PO traceable to its exact scoring inputs.

Scenario-Level Walkthroughs — Mock Environment

Three scenarios from the live mock environment showing the concrete before/after difference in how each situation is handled. All data drawn from scenarios/ and mock_data/ in the Betsy repository.

Scenario 1 — Stockout Warning: SKU-003

Steel Rods 10mm | Stock: 45 units | Daily usage: 30 | Days remaining: 1.5

BEFORE (MANUAL)

1Monday morning ERP check

Jenny spots 45 units during her periodic scan. Production line shuts down Wednesday morning unless stock arrives in time.

2Emergency email to 3 suppliers

Jenny contacts FastParts (2-day), BulkSupply (7-day), and QuickShip (1-day). ~30 min drafting

3Wait 24-48 hrs for replies

With only 1.5 days of stock, even QuickShip's 1-day lead is dangerously close. Approval chain adds another day. Production line at risk

AFTER (BETSY)

1Continuous monitor detects breach

Betsy sees SKU-003 at 45 units vs reorder_point of 200. days_remaining = 45 ÷ 30 = 1.5. Alert fires immediately — no human needed. <1 min detection

2All 6 suppliers queried, QuickShip selected

Only QuickShip Express (SUP-003) offers 1-day lead for SKU-003. Selected automatically by scoring. PO generated. <5 sec sourcing

3PO dispatched, Jenny notified

PO sent immediately. Stock arrives next day — 0.5 days buffer before shutdown. Jenny receives dashboard update. Expected supplier: QuickShip Express

Scenario 2 — Duplicate Invoice: INV-2024-099

Supplier: MidRange Supply (SUP-006) | Amount: \$1,470 | SKU-004 ×10

BEFORE (MANUAL)

1Finance receives INV-2024-099

Third \$1,470 invoice from MidRange Supply for 10× Circuit Board Type A. Looks legitimate at first glance.

2Manual spreadsheet cross-check

With 20+ invoices in the system, checking for duplicates manually takes 30-60 minutes and is error-prone. 30-60 min manual effort

3Duplicate likely paid

INV-2024-089 and INV-2024-091 were already in the system undetected. INV-2024-099 likely processed and paid. \$1,470 at risk

AFTER (BETSY)

1Invoice submitted to /api/invoices

_find_duplicates() runs automatically on every POST. No manual trigger required. Algorithm checks all invoice pairs.

2Match detected: same supplier, same amount, 29 days apart

INV-2024-099 matches INV-2024-089: SUP-006, \$1,470.00, 29-day gap → HIGH RISK (threshold: 30 days). <100ms detection

3

Flagged before payment
duplicate_flags array returned in API response. Invoice held for Jenny review. Payment blocked.

\$1,470 protected

Scenario 3 — Price Spike Escalation: SKU-003
Best quote: \$13.60 (ValueFirst) vs historical avg \$11.40 (+19.3%)

BEFORE (MANUAL)

- 1
- All quotes arrive elevated**
FastParts: \$20, BulkSupply: \$15.70, QuickShip: \$24, PrecisionParts: \$17.90, ValueFirst: \$13.60. All 60–110% above normal.
- 2
- No formal price baseline**
Jenny knows "it seems high" but has no threshold to enforce. Decision is instinct-based under production pressure.

No guardrail
- 3
- Risk of overpaying under pressure**
Jenny may accept ValueFirst's \$13.60 (best available) without realising it is 19.3% above the historical avg of \$11.40.

Budget overrun risk

AFTER (BETSY)

- 1
- Best quote vs stored historical avg compared**
historical_avg_price for SKU-003 = \$11.40. Best current: \$13.60 from ValueFirst (10-day lead). Deviation: +19.3%.
- 2
- Escalation triggered — human decision required**
Spike exceeds monitoring threshold. Jenny receives full context: all 5 quotes, historical avg, recommended action, reasoning.

Guardrail active
- 3
- Informed decision — not reactive**
Jenny can accept market price, delay reorder, or negotiate directly. Decision is data-driven, not made under blind pressure.

Expected action: flag_for_approval

Weekly Time-Motion Analysis

BEFORE — Jenny's Weekly Procurement Load

Inventory monitoring (manual ERP checks)	4-5 hrs
Supplier RFQ emails & responses	6-8 hrs
Quote comparison & supplier selection	2-3 hrs
PO drafting & approval chasing	4-5 hrs
Delivery tracking & supplier follow-up calls	5-8 hrs
Invoice matching & discrepancy resolution	3-4 hrs
Monday morning reactive firefighting	4-6 hrs
Total Weekly Procurement Load	30-39 hrs

AFTER — Jenny's Weekly Procurement Load (With Betsy)

Inventory monitoring	0 hrs (Betsy)
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Supplier RFQ & sourcing	0 hrs (Betsy)
Routine PO creation	0 hrs (Betsy)
High-value / anomaly PO review (1-click)	0.5-1 hr
Delivery tracking	0 hrs (Betsy)
Invoice exception review (flagged items only)	0.5-1 hr
Strategic dashboard & supplier development	1-2 hrs
Total Weekly Procurement Load	2-4 hrs

Net time reclaimed: ~27-35 hours per week. Redirected to strategic supplier relationship management, contract negotiation, multi-site procurement strategy, and cross-functional projects — the work Jenny was hired for, not the operational firefighting she is currently trapped in. The agent does not replace Jenny's judgment; it reserves it for decisions that actually require it.